So the reality of a quantum system, according to austere quantum mechanics, is described by a wave function or quantum state, which can be thought of as a superposition of every possible outcome of some observation we might want to make. How do we get from there to the annoying reality that wave functions appear to collapse when we make such measurements?

Start by examining the statement “we measure the position of the electron” a little more carefully. What does this measurement process actually involve? Presumably some lab equipment and a bit of experimental dexterity, but we don’t need to worry about specifics. All we need to know is that there is some measuring apparatus (a camera or whatever) that somehow interacts with the electron, and then lets us read off where the electron was seen.

In the textbook quantum recipe, that’s as much insight as we would ever get. Some of the people who pioneered this approach, including Niels Bohr and Werner Heisenberg, would go a little bit further, making explicit the idea that the measuring apparatus should be thought of as a classical object, even if the electron it was observing was quantum-mechanical. This line of division between the parts of the world that should be treated using quantum versus classical descriptions is sometimes called the *Heisenberg cut.* Rather than accepting that quantum mechanics is fundamental and classical mechanics is just a good approximation to it in appropriate circumstances, textbook quantum mechanics puts the classical world at center stage, as the right way to talk about people and cameras and other macroscopic things that interact with microscopic quantum systems.